Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/05/2005
PI Name:	Ronney, Paul D. Sc.D.		
Project Title:	Radiative Enhancement Effects on Flame Spread (REEFS): A Space Flight Experiment Proposal		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ronney@usc.edu	**Fax:**	FY 213-740-8071
PI Organization Type:	UNIVERSITY	**Phone:**	213-740-0490
Organization Name:	University of Southern California		
PI Address 1:	Department of Aerospace and Mechanical Engineering		
PI Address 2:	OHE 430J University of Southern California		
PI Web Page:	http://carambola.usc.edu		
City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	04/27/2003	**End Date:**	01/03/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Gokoglu, Suleyman A	**Contact Phone:**	216-433-5499
Contact Email:	Suleyman.A.Gokoglu@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1056		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	The research is divided into three topics: 1) radiation-dominated flame spread over thick fuel beds, 2) concurrent-flow flame spread rates and extinction conditions effects of convective and radiative losses, and 3) addition of gaseous fuels to the oxidizing atmosphere. Flame spread over flat solid fuel beds is a useful means of understanding more complex, two-phase, non-premixed flames, such as those found in fires in enclosures, e.g., manned spacecraft and terrestrial buildings. Both transport and chemical effects are well known to affect the characteristics of flame spread. Conductive, convective and radiative transport processes may couple to each other and to chemical effects in unusual ways. Consequently, the goal of this work is to extend previous studies by employing μg experiments with emphasis on the atmospheres and flow environments likely to be present in fires which might occur at μg. The goal of this work is to provide information that will enhance crew safety in manned spacecraft including the Space Shuttle, International Space		

Task Description:	Station, and potential future missions to the moon and Mars. The understanding and control of accidental fires is a critical safety issue in both terrestrial and space-borne environments. Space environments are different from earth-based environments for several reasons. Of course, the major difference is the lack of buoyancy-induced convection. Another difference is that fires in spacecraft may occur in non-standard atmospheres because of the use of non-standard breathing atmospheres or the use of diluents as fire extinguishing agents. Consequently, the understanding of these effects is critical to understanding how fires might start, spread, and be extinguished at µg conditions. One of the tasks of this project, the study of CO2 as a fire extinguishing agent, is particularly relevant to fire safety on the International Space Station, whose fire suppression system uses CO2 as the extinguishing agent. Our experiments show that CO2 is less effective that other agents such as helium at microgravity, whereas the opposite is true at earth gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radiation is the main mechanism of fire spread between buildings, thus understanding radiation effects on fires is useful to fire prevention and suppression on earth as well as in space.
Task Progress:	Earth gravity and drop-tower experiments have shown that for thermally-thick foam fuel beds, three fire extinguishing agents (helium, nitrogen and carbon dioxide) have nearly the same efficacy in reducing flame spread rates and causing flame extinction for a wide range of total pressures and oxygen mole fractions. Moreover, even at microgravity, for higher total pressures and lower oxygen mole fractions, all three agents have nearly the same efficacy in reducing flame spread rates and causing flame extinction. However, at microgravity, for lower total pressures and higher oxygen mole fractions, CO2 is clearly the least effective of these three agents tested and He is the most effective. The spread rates in undiluted atmospheres is higher at microgravity, but dilution has a stronger effect on spread rates at microgravity than at earth gravity. On a simple heat capacity basis, one would expect CO2 to be the most effective extinguishing agent and He the least effective since CO2 has the highest heat capacity of the three extinguishing agents tested and He has the lowest. Thus, dilution by CO2 should lead to the largest reduction in flame temperature and thus the smallest spread rate. In contrast, the experimentally observed results show exactly the opposite trend at low pressures and high oxygen mole fractions. We believe there are three reasons for this behavior. First, the Lewis number of O2 in He is much higher than O2 than in N2 or CO2 (1.20 vs. 1.00 vs. 0.84), which, as previous work has shown, leads to significantly lower flame temperatures and lower spread rates for He than would be expected on a simple heat-capacity basis. Second, the dramatically higher thermal conductivity and thermal diffusivity of He compared to N2 or CO2 (1.82 vs. 0.22 vs. 0.12 cm2/s) leads to much thicker flames and thus greater radiative loss for the same spread rate in the case of He. Third, unlike CO2, N2 and He are radiatively non-participating and thus no reabsorption or re-emission occurs. The reabsorption reduces heat loss to ambient and the re-emission increases the total heat flux to the fuel bed, both of which increase the spread rate in the case of CO2. There are also other potential advantages to He as an extinguishing agent for spacecraft applications. For example, He is far more effective on a mass basis at both 1g and µg. Additionally, He has no physiological activity, unlike CO2 which affects human respiration. Moreover, as compared to N2 or CO2, He is not very soluble in water (about 8.6 ml/liter at 20°C, compared to 16.9 for N2 and 1019 for CO2) and thus has less tendency to cause bloodstream bubble formation following accidental or intentional spacecraft cabin depressurization. (The Space Shuttle uses intentional partial cabin depressurization, with a corresponding increase in O2 mole fraction, as a standard procedure for Extra-Vehicular Activity because it leads to a shorter period of time that the EVA astronauts need to pre-breathe pure O2 before suiting up and depressurizing to the 0.2 atm, pure O2 environment of the EVA suits.) The effect of opposed flow velocity (U) on flame spread rate was also measure for foam fuels. With 35% O2 in CO2 at 2 atm total pressure, the spread rate shows a slight minimum at an intermediate value of U (about 1 cm/sec). A minimum spread rate at intermediate U is consistent with the predictions of a simple model we developed. The theoretical predictions show that for denser fuels, the differences in spread rate between quiescent microgravity, microgravity with weak flow and earth gravity are more significant, indicating that for denser fuels, thus earth gravity data may be even more atypical of spread rate characteristics at quiescent or low-flow microgravity conditions.
Bibliography Type:	Description: (Last Updated: 01/29/2007)
Abstracts for Journals and Proceedings	Son, Y.; Zouein, G.; Gokoglu, S.; Ronney, P. D. "COMPARISON OF CARBON DIOXIDE AND HELIUM AS FIRE EXTINGUISHING AGENTS FOR SPACECRAFT" Proceedings of the 4th Joint meeting, U. S. Sections, Combustion Institute, Philadelphia, PA N/A , Mar-2005